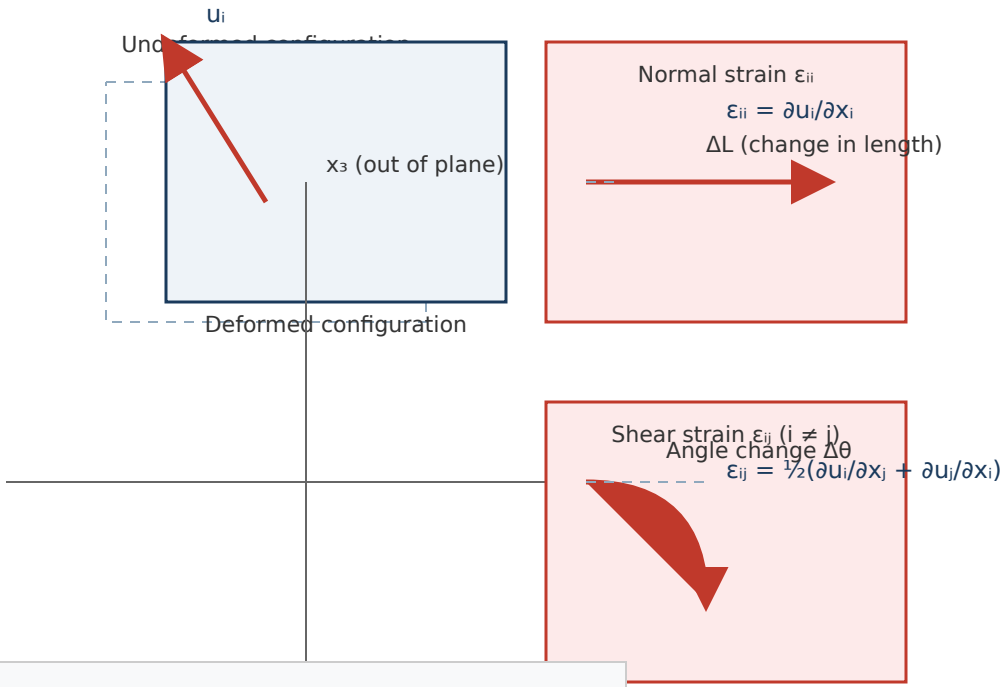
Fundamental continuum kinematics

Displacement, strain, rotation, and velocity in an infinitesimal continuum element



Normal strain ε_{ii}

$\varepsilon_{ii} = \partial u_i / \partial x_i$

ΔL (change in length)

Shear strain ε_{ij} $(i \neq j)$

Angle change $\Delta\theta$

$\varepsilon_{ij} = \frac{1}{2}(\partial u_i / \partial x_j + \partial u_j / \partial x_i)$

Rotation tensor ω_{ij}

$\omega_{ij} = \frac{1}{2}(\partial u_i / \partial x_j - \partial u_j / \partial x_i)$

Rigid-body rotation

Velocity field v_i

$v_i = \partial u_i / \partial t = \dot{u}_i$

Material time derivative

Key definitions:

$u_{i,j} = \varepsilon_{ij} + \omega_{ij}$ (displacement gradient decomposition)

$\varepsilon_{ij} = \frac{1}{2}(u_{i,j} + u_{j,i})$ (infinitesimal strain)

$\omega_{ij} = \frac{1}{2}(u_{i,j} - u_{j,i})$ (rotation tensor)

$v_i = \dot{u}_i$ (velocity)

Infinitesimal strain assumption: $\|\varepsilon_{ij}\| \ll 1$. Valid for linearized wave propagation.